MILITARY SPECIFICATION

CARTRIDGE, 40MM, HE, M386

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of cartridge designated as Cartridge, 40MM, HE, M386.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-P-20444	-	Primer, M42
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8835951	Cartridge, 40MM, HE, M386

PUBLICATIONS

ARMY

ET-8835951	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	Transportation, Parts 171-179

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(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 171-179 (latest revision)").

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. -The cartridge shall complv with all requirements specified on Drawing (Dwg) 8835951, all associated drawings and with all requirements specified in applicable specifications.

3.3 Loading of high explosive.

3.3.1 Each ball and skirt assembly just before loading shall be inspected for presence of primer paint and any foreign matter. Ball and skirt assemblies with the interior wall incompletely coated or not dry shall not be loaded.

3.3.2 Solid explosive shall not be present before loading or added during pouring or cooling.

3.3.3 The fuze seating surface shall be free of caked explosive. A chemical cleaner, if used, shall not contaminate the explosive.

3.3.4 Clean scrap produced in regular operations may be remelted and used, provided it complies with the grade Of explosive specified except for form and viscosity. Clean scrap is defined as scrap formed in kettles, pails, riser, and loading machines used in pouring, casting and drilling operations.

3.4 Air pressure.-The cartridge assembly shall withstand an air pressure of three (3) pounds-per-square-inch (psi) for five (5) seconds minimum, without a drop in pressure when tested as specified in 4.4.7.

3.5 Primer functioning.-The primer shall function without evidence of squibbing or hangfire when tested as specified in 4.4.11, Para 6.6 of MIL-P-20444B, Amendment 3, contains definitions of functioning failures.

3.6 X-ray examinations of fuze assemblies prior to assembling to ball and skirt assembly.-The fuze assembly shall be X-rayed for improperly assembled, missing parts, armed or partially armed (as evidenced by the detonator moving any degree from the safe position) when tested as specified in 4.4.1.

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3.7 Production control check on loaded M118 cartridge cases.-There shall be no evidence of a primer malfunction when tested as specified in 4.4.2.

3.8 X-ray examination of ballistic samples.-Prior to forwarding the cartridges to the proving ground for ballistic testing, as specified in 4.3.3.15 and 4.3.3.16, they shall be subjected to X-ray examination when tested as specified in 4.4.3.

3,9 Functioning.

3.9.1 Cartridge.-The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 247 plus or minus (±) 7 feet-per-second and a standard deviation not exceeding 4.5 feet-per-second when tested as specified in 4.4.10.

3.9.2 Safety.-There shall be no premature burst (See 6.7) or evidence thereof in any of the test firings when tested as specified in 4.4.10.

3.10 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.11 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.12).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and *services* conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product. -At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commerical specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent’s authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.-Not applicable.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term “inspection lot” as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine a conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes for process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification

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shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Cartridge case metal parts from one interfix lot number from one manufacturer.
b. Ball and skirt assemblies from one interfix lot number from one manufacturer.
c. Composition B from one interfix lot number from one manufacturer.
d. Fuzes from no more than two (2) consecutively produced serial lots from one manufacturer.
e. Primers from one lot number from one manufacturer.
f. Propellant from one lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cartridge case assembly, prior to assembling to projectile and prior to inserting base plug and primer (See Dwg 8844609 covering a detail of Dwg 8835951).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Powder charge cup missing	Gage	01001
2.	Spilled powder from cartridge case	Visual	01002
3.	Propellant charge weight obviously incorrect (see paragraph 6.10)	Visual	01003

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Major	100 Percent Inspection		
101.	Propellant charge weight obviously incorrect (See 6.10)	Visual	01004

3

Minor: None defined

4.3.2.2 Cartridge case loading assembly, prior to assembling to projectile (See Dwg 8844609 covering a detail of Dwg 8835951).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush with respect to base of case	Gage	02001
Major	AQL 1.00 Percent		
101.	Base plug above flush	Gage	02002
102.	Depth of base plug, maximum (max)	Gage	02003
103.	Primer above flush or more than .005 below with respect to base plug	Gage	02004
104.	Primer damaged	Visual	02005
105.	Excessive varnish on exterior surface of primer	Visual	02006
Minor	AQL 1.00 Percent		
201.	Marking misleading or unidentifiable	Visual	02007
202.	Evidence of poor workmanship (See 3.10)	Visual	02008

4.3.2.3 Loading assembly, prior to loading (See Dwg 8840503 covering a detail of Dwg 8835948).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor	AQL 0.65 Percent		
201.	Foreign matter on interior surface (See 3.10)	Visual	03001

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4.3.2.4 Loading assembly (See Dwg 8840503 covering a detail of Dwg 8835948).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Excessive cavitation on bottom of fuze cavity	Visual	04001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	04002

4.3.2.5 Projectile, prior to assembling fuze and prior to assembling sealing compound on fuze threads (See Dwg 8835948 covering a detail of Dwg 8835951).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
1Q1.	Distance from seating surface on fuze to end of booster cup, max	Gage	05001
102.	Gap from fuze threads to booster cup, missing	Visual	05002
Minor:	None defined		

4.3.2.6 Projectile, prior to assembling ogive (See Dwg 8835948 *covering* a detail of Dwg 8835951).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Fuze torque failure	Test	06001
102.	Length from curved surface to top of fuze	Gage	06002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	06003

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4.3.2.7 Projectile (See Dwg 0835940 covering a detail of Dwg 8835951).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Weight incorrect	Balance	07001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	07002

4.3,2.8 Cartridge, 40MM, HE, M386 (See Dwg 8835951).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Chamber gage failure (after 5 hr min cure time at 70°F min in a nose down position)	Gage	08001
Major:	None defined		
Minor	AQL 1.50 Percent		
201.	Total length	Gage	08002
202.	Marking missing	Visual	08003
203.	Evidence of poor workmanship (See 3.10)	Visual	08004

4.3.2.9 Unsealed fiber box (See Dwg 8835105).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Improper number of cartridges	Visual	09001
102.	Separator missing	Visual	09002
Minor:	None defined		

4.3.2.10 Sealed fiber box, prior to sealing bag (See Dwg 8835105).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major	AQL 0.40 Percent		
101.	Container damaged	Visual	10001
102.	Corners not blunted	Visual	10002
Minor	AQL 1.00 Percent		
201.	Contents loose	Manual	10003
202.	Marking misleading or unidentifiable	Visual	10004

4.3.2.11 Unsealed wooden packing box (See Dwg 8835104).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101*	Marking of bag misleading or unidentifiable	Visual	1100l
102.	Bag torn or perforated	Visual	11002
103.	Bag improperly sealed	Visual	11003
Minor:	None defined		

4.3.2.12 Sealed wooden packing box (See Dwg 8835104).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Box damaged	Visual	12001
Minor	AQL 1.50 Percent		
201.	Contents loose	Manual	12002
202 l	Car seal missing or improper positioned	Visual	12003
203.	Marking misleading or unidentifiable	Visual	12004

4.3.3 Testing.

4.3.3.1 Check test for deterioration of primers (See MIL-P-20444) Major Defect.-If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two (2) years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must l

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satisfactorily pass, the check test for deterioration specified in MIL-P-20444. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into cartridge (See 6.4) Code No. 13001.

4.3.3.2 Holding fixture vent check, Code No. 14001.-The cartridge case holding fixture used in the base plug insertion operation shall have its air vents check hourly for any obstruction. If a vent is found to be closed or clogged, the vent shall be cleared and the previous hour's production shall be X-rayed 100 percent to ensure that the propellant charge cups have not been damaged. Cartridge cases found to have damaged charge cups shall be classed defective and removed from the lot.

4.3.3.3 Propellant weight of cartridge case, Code No. 15001.-The propellant weight shall be determined and then check weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing or determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirement specified on Dwg 8844609 shall be classified defective and removed from the lot.

4.3.3.4 Propellant weight of cartridge case, alternate method, Code No. 16001.-At the start of production, three hundred eighty (380) consecutive samples per station, per machine shall be selected and weighed 100 percent. All samples shall comply with the applicable drawing requirement. If all samples comply with the drawing requirement, five (5) samples, per station, Per every two (2) hours production shall be selected and the average weight of the five (5) samples must meet the assessed mean propellant weight within + 4 milligrams. If the average weight fails the requirement, correction will be made and three hundred eighty (380) consecutive samples weighed from that station. If all meet the requirement, the sampling every two (2) hours will resume.

4.3.3.5 X-ray examination of fuze assemblies prior to assembling to ball and skirt assembly (See Para 3.6) Code No. 17001.-Each fuze assembly, prior to assembling to ball and skirt assembly, shall be subjected to this test. If any fuze assembly shows evidence of being improperly assembled, having missing parts, being armed or partially armed, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

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4.3.3.6 Production control check on loaded M118 cartridge case (See Para 3.7) Major Defect, Code No. 18001.-Five (5) loaded M118 cases shall be randomly selected from every two (2) hours production from each primer assembly machine. The loaded M118 cases produced by each primer-assembly machine shall be kept segregated and identified. They shall not be used until this test has been performed. If any sample fails to comply with the applicable requirement, the two hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.7 X-ray examination of ballistic samples (See 3.8) Code No. 19001.-Prior to forwarding the cartridges to the proving ground for ballistic testing, as specified in 4.3.3.15 and 4.3.3.16, they shall be subjected to X-ray examination. The cartridges shall be X-rayed for critical and major defects. If any critical defect is found, the lot shall be rejected including the ballistic samples. If any major defect is found, it shall be noted and the samples forwarded to the proving ground. There shall be no replacement of samples. This test shall be performed as specified In 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.8 Security of projectile (See Dwg 8835951) Major Defect, Code No. 20001.-After assembly of the cartridge case to the projectile, eighty (80) cartridges shall be selected from each lot and subjected to this test. If four (4) or more fail to comply with the drawing requirement, the lot shall be rejected. In addition, if any one is less than 50 pounds, the lot shall be rejected. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.9 Security of crimp of base plug on cartridge case assembly (See Dwg 8844609) Major Defect, Code No. 21001.-Twenty (20) each of cartridge case, base plug, and powder charge cup less propellant shall be selected from each lot and subjected to this test. If the crimped area of one or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4.

4.3.3.10 Security of crimp of ogive of projectile (See Dug 8835948) Major Defect, Code No. 22001.-Twenty (20) each of ogive and empty ball and skirt assemblages shall be selected from each lot. The ogive shall be cemented and crimped in place and subjected to this test. If the ogive of two (2) or more assemblies fail to comply with the minimum requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3,4. (when applicable).

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4.3.3.11 Heat seal test of packing seals (See Dwg 8835105, Sheet 2) Major Defect, Code No. 23001.-The samples shall be selected in accordance with MIL-P-116. Rejection shall be in accordance with MIL-P-116. The test shall be performed as specified in 4.4.8.

4.3.3.12 Specific gravity of the charge of ball and skirt loading assembly (See Dwg 8840503) Major Defect, Code No. 24001.-Two (2) sample cup assemblies shall be selected from each four (4) hour's production and subjected to this test. If any sample fails to comply with the specific gravity requirements specified on the applicable drawing, the four (4) hour's production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.9.

4.3.3.13 Air pressure (See para 3.4) Major Defect, Code No. 25001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter L, Single Plan, with an AQL of 1.50 percent. The test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4.

4.3.3.14 Primer functioning (See Para 3.5) Major Defect, Code No. 26001.-Sampling shall be in accordance with Paragraph 4.2.3.2.2 of MIL-P-20444B, except that the sample size shall be eight hundred (800) primers from each lot. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected. primers not originally purchased to this requirement shall be tested for the requirement of Paragraph 3.5. The test shall be performed as specified in 4.4.11.

4.3.3.15 Functioning (See Para 3.9) Major Defect, Code No. 27001.-Beginning with the first lot produced and continuing until three (3) acceptable lots have been accepted under the rejection criteria specified in 4.3.3.15.1, the Government inspector shall select two hundred eighty-eight (288) cartridges from each lot for this test.

4.3.3.15.1 Rejection.-The lot shall be rejected if:

a. Twelve (12) or more fail high order detonation at initial impact (See 6.8). (Major Defect)

b. Any primer blow backs (as evidenced by perforation of the cup and extrusion of the primer around firing pin causing difficulty in opening weapon). (Major Defect)

co Any premature burst occurs. (Critical Defect)

d. Mean velocity and/or standard deviation fail to meet the applicable requirement. (Major Defect)

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e. Any primer misfires. (Major Defect)

f. (1) Thirty (30) or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major Defect)

(2) One (1) or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor. (Major Defect)

g. Any projectile sticks in gun bore or fails to attain 60 percent of the mean uncorrected velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity, shall be considered as a failure. (Critical Defect)

h. Four (4) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major Defect)

i. Any metal part separation occurs prior to impact. (Critical Defect)

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on a retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

4.3.3.15.2 Initial production test plan, (informational test only) Code No. 27002.-Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) acceptable lots that are accepted in accordance with Paragraph 4.3.3.15, an extra. sample of ninety (90) rounds shall be selected from each lot and pre-conditioned and tested as follows:

a. The ninety (90) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

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c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds +165 degrees F ± 5 degrees F.
30 rounds ambient (between + 40 and + 110 degrees F).
30 rounds -65 degrees F ± 5 degrees F.

d. Fire the rounds in accordance with Paragraph 4.4.10 and with TECOM Material Test procedure, TECP 700-700 Volume 4. Test results, failure analysis reports, and other information, as requested shall be forwarded to Picatinny Arsenal, ATTN: SMUPA-ND3 for evaluation.

4.3.3.16 Functioning (See Para 3.9) Major Defect, Regulation Production Code No. 28001.-After three (3) acceptable lots have met the criteria of 4.3.3.15, the Government inspector shall select seventy-two (72) cartridges from each lot for this test. If two (2) or more lots fail the criteria of 4.3.3.16.1, the consecutive lots thereafter shall revert to the criteria of 4.3.3.15,

4.3.3.16.1 Rejection. -The lot shall be rejected if:

4 or more fail high order detonation of initial impact (See 6.8). (Major Defect)

b. Any primer blow backs (as evidenced by perforation of the cup and extrusion of the primer cup around firing pin causing difficulty in opening weapon). (Major Defect)

c. Any premature burst occurs. (Critical Defect)

d. Mean velocity and/or standard deviation fail to meet the applicable requirement. (Major Defect)

e. Any primer misfires. (Major Defect)

f. (1) 5 or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major Defect)

(2) 1 or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor. (Major Defect)

g. Any projectile sticks in gun bore or fails to attain 60 percent of the mean uncorrected velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered a failure.) (Critical Defect)

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h. 3 or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major Defect)

i. Any metal part separation occurs prior to impact. (Critical Defect)

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, then sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and the mean velocity of the control lot from that previously obtained for the control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

4.3.4 Inspection equipment. -Equipment Tabulation (ET) number 8835951 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 X-ray examination of fuze assembly prior to assembling the ball and skirt assembly.-Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by X-ray equipment or any other method satisfactory to the Contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot. This test shall be conducted in two (2) phases:

a. From the top of the fuze assembly down and (b) from the side.

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4.4.2 Production control check on loaded M118 cartridge cases.-The Loaded M118 cartridge case shall be fired from an approved launcher placed on an approved mount. Observation shall be made for primer functioning. This test is a destructive test, Parts so tested shall not be returned to the lot. NOTE : No projectile is required for this test.

4.4.3 X-ray examination of ballistic samples.-The cartridge, prior to forwarding to the proving ground, shall be subjected to X-ray examination using the X-ray equipment specified on the Equipment List or any other method satisfactory to the Contracting Offocer. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Security of projectile.-The cartridge shall be placed in the fixture shown on the Equipment List and the axial force specified on Dwg 8835951 shall be applied. Projectile must separate from the cartridge case in the range specified on Dug 8835951. This test is a non-destructive test. Parts so tested may be returned to the lot. Stake shall be removed from loaded case prior to returning to the lot.

4.4.5 Security of crimp of base plug on cartridge case assembly.-The assembly shall be placed in the fixture on the Equipment List and the minimum unseating force specified on Dug 8844609 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot l

4.4.6 Security of crimp of ogive of projectile.-The assembly shall be placed in the fixture shown on the Equipment List and the minimum unseating force specified on Dug 8835948 shall be applied until failure of the crimp area of the ogive occurs. This test is a destructive test. Parts so tested shall not be returned to the lot. (When applicable).

4.4.7 Air pressure.-The cartridge shall be placed in the fixture shown on the Equipment List and a measured quantity of air applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.8 Heat seal test of packing seals.-This test shall be performed in accordance with MIL-P-116. Seals so tested shall not be returned to the lot .

4.4.9 Specific gravity of the charge of ball and skirt loading assembly .-The specific gravity of the composition B shall be determined as follows:

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Weigh the empty ball and skirt assembly in grams and record as We. Fill the empty assembly with water at 75 degrees ± 2 degrees F, weigh in grams and record the weight as Ww. Determine the volume by subtracting We from Ww.

Load the assembly with Composition B, drill the fuze well cavity, weigh in grams and record the weight as W1. Determine the charge weight by Subtracting We from W1. Calculate the specific gravity as follows:

$$\text{Specific gravity in grams/CM} = \frac{W1 - We}{Ww - We} - 3.21$$

4.4.10 Functioning.-This test shall be conducted at a Government owned proving ground. The cartridge shall be fired from an approved launcher placed on an approved mount to impact at 200 meters minimum from the launcher on a six inch thick loose gravel pad. The gravel used shall have a 1/2 inch maximum size. The velocity shall be measured at 20 feet from the muzzle of the launcher using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. NOTE : To insure that the chronographs display accurate time intervals, the ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more than plus or minus 0.005, the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation. (No more than four (4) rounds in a sample size of 72 or twelve (12) rounds in a sample size of 288 may be disregarded. If the applicable number of improper ratios is exceeded, additional round(s) must be fired). Observation shall be made and results recorded for functioning, mean velocity and standard deviation (See 6.6). The mean velocity of the lot being tested shall be corrected by the correction factor determined for the particular test launcher being used on the day of the test firing by firing control rounds in accordance with Table I. When velocity values are suspected to be outliers, they shall be tested, using the statistical procedure approved by the Government proving ground and the mean velocity and standard deviation adjusted accordingly. However, no more than four values consisting of no more than two on the high side and no more than two on the low side, shall be considered for exclusion.

TABLE I (See 6.11)

Standard Deviation for Control Lots	No. of Rounds
3 or less	24
3.1 - 3.5	36
3.6 - 4.0	60
4.1 - 4.5	84

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4.4.10.1 When any one sample lot from a regular ammunition lot is broken into two (2) groups and fired separately, the following procedure shall be followed when calculating the mean velocity and standard deviation:

1. Individual velocities shall be corrected for each group, using the correction factor of that group.

2. The mean velocity and standard deviation shall be calculated from all the corrected velocities.

4.4.10.2 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.11 Primer functioning.-Each primer of the 800 samples shall be tested as specified in MIL-P-20444B, paragraph 4.3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with Dwg 8835105.

5.2 Packing.

5.2.1 Level A.-The cartridge, preserved and packaged as specified in 5.1.1, shall be packed in accordance with Dug 8835104.

5.3 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

5.4 Marking.-Marking shall be in accordance with Dug 8835104 and 8835105, and Code of Federal Regulations 49, Title, Parts 171-179.

6. NOTES

6.1 Ordering data.-procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

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6.2 Inspection code numbers. -The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment. -The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists. -Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment design, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to' prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment . Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes. -The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
Mu - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs. -Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design. -All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test. -The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc, permitted while the primer assembly lot was under the jurisdiction of either contractor.

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6.5 Submission of test data. -In addition to the normal distribution of records, when the cartridges are procured by the US Army Munitions Command one (1) copy of all ballistic data and ammunition data cards sha11 be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.6 Standard deviation shall be calculated from the following formula or other approved formula:

$$S = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}}$$

Where:

X = each individual value

$\bar{X}$ = sample arithmetic mean $\frac{\Sigma X}{n}$

n = sample size

$\Sigma (X - \bar{X})^2$ = the sum of the squares of the differences between the sample mean and each individual value

n - 1 = one less than the sample

6.6.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used:

$$S = \sqrt{\frac{n \Sigma x^2 - (\Sigma X)^2}{n (n-1)}}$$

6.7 Premature burst .-A premature burst is considered to have occurred if the round functions in gun bore or in flight.

6.8 High order functioning.-Evidence of high-order functioning shall be proper disintegrated of the ball as determined by the proof director.

6.9 Combining of proving ground tests. -When the contractor for the cartridge is also contractor for one or more of the components thereof the proving ground tests of the contractor may be combined with the proving ground tests of the cartridge, to save expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all of the proving ground tests specified for the Componont, the addition tests specified in the component specification shall be conducted.

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6.10 If any cartridge case, prior to insertion of the base plug and primer, is suspected of containing an underweight or overweight propellant charge, based on visual inspection, it will be removed from the lot and weighed on suitable scales. Any cartridge case found with less than 75% or more than 125% of the assessed propellant load will be classified as a Critical defect and removed from the lot.

Any cartridge case found to have a propellant weight out of drawing tolerance with within + 25% of the assessed propellant load will be classified as a Major defect and removed from the lot.

6.11 Table shall be used at the loading plants for powder assessment.

6.12 Submission of results of contractor-conducted examinations and tests. -Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.13 Proving ground test.- Initial production testing shall be conducted in accordance with US Army Test and Evaluation Command Initial Production Test Plan No. CART-40GL-VAR.

6.14 Proving Ground Test Summary

TEST	SAMPLE SIZE	REQUIREMENT
FUNCTIONING		
First 3 Lots	288	See 3.9 See 4.3.3.15.1
IPTP	270	See 4.3.3.15.2
Regular Production	72	See 3.9 See 4.3.3.16.1

6.15 Supersession data. -This specification includes the requirements of , and supersedes, Purchase Specification AEI-61D, dated 23 September 1968.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1310-A-120

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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